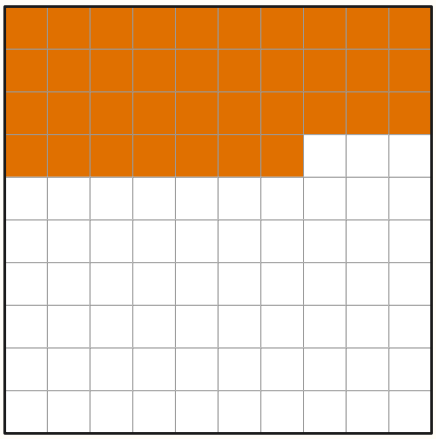
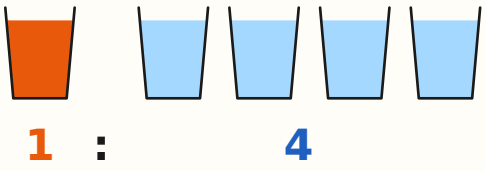
Chapter 4

Percentage and ratio

comparing amounts: out of a hundred, and parts of a recipe



37 out of 100 = 37%



squash to water 1 : 4

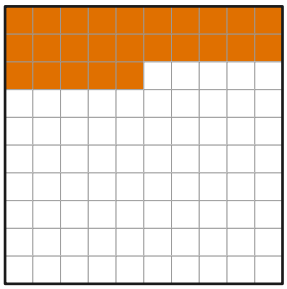
"20% off". "Mix 1 part squash with 4 parts water". "The model is 1 : 72 scale". Both **percentages** and **ratios** answer the same kind of question: how big is one amount compared to another? They're fractions (Chapter 2) dressed for everyday life.

This chapter goes through all of Lecture 3: percentages of amounts, going up and down by a percentage, working backwards, ratios, sharing in a ratio, scaling with the "one of them" table, and mixing solutions.

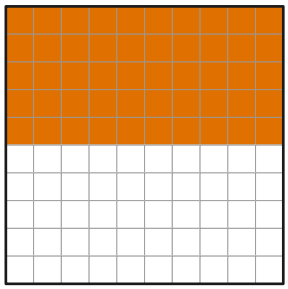
Goes with **Lecture 3** (ratios and proportions).

Per cent means "out of a hundred"

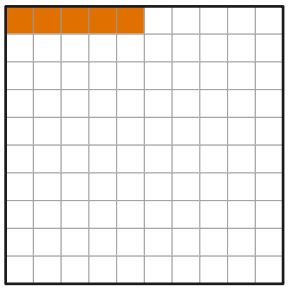
"Cent" means hundred (a century is 100 years). So **per cent** means **per hundred**: chop the whole thing, whatever it is, into 100 equal pieces, and count pieces. The symbol **%** is a tiny drawing of "/100" (a slash between two zeros).



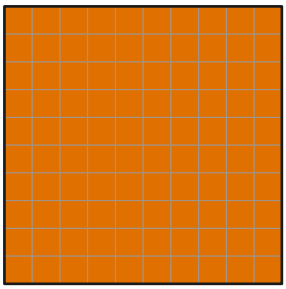
25% = 1/4



50% = 1/2

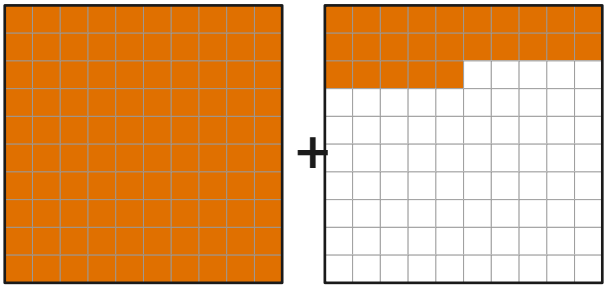


5% = 1/20



100% = all of it

More than 100% just means more than one whole. **125%** is one full square and another quarter of a square:



125% = 125/100 = 1.25

(one and a quarter wholes)

Three outfits for the same number

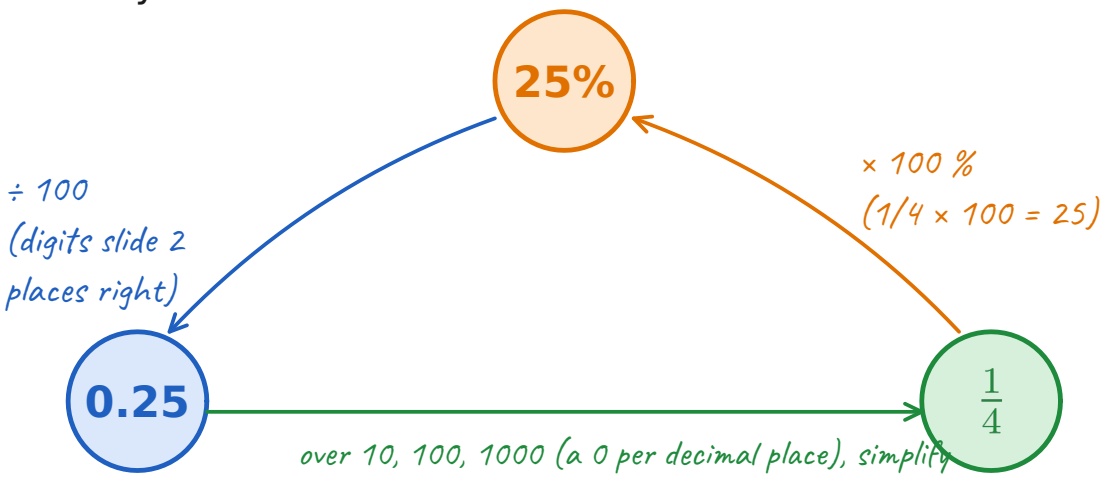
RULE: converting percentages, decimals and fractions

Percentage to decimal: divide by 100. $35\% = 0.35$.

Percentage to fraction: write over 100 and simplify. $35\% = \frac{35}{100} = \frac{7}{20}$.

Decimal or fraction to percentage: multiply by 100. $0.07 = 7\%$; $\frac{3}{8} \times 100 = 37.5\%$.

A quarter can wear three outfits: $\frac{1}{4}$, 0.25 and 25%. Moving between them is always one of these arrows:



Your Lecture 3 examples, one arrow each:

$$5\% = \frac{5}{100} = 0.05 \longleftarrow \text{percent} \rightarrow \text{divide by 100}$$

$$125\% = \frac{125}{100} = 1.25 \longleftarrow \text{same arrow, bigger number}$$

$$0.75 = \frac{75}{100} = 75\% \longleftarrow \text{decimal} \rightarrow \text{multiply by 100}$$

$$1.583 = 158.3\% \longleftarrow \text{decimal} \rightarrow \text{multiply by 100}$$

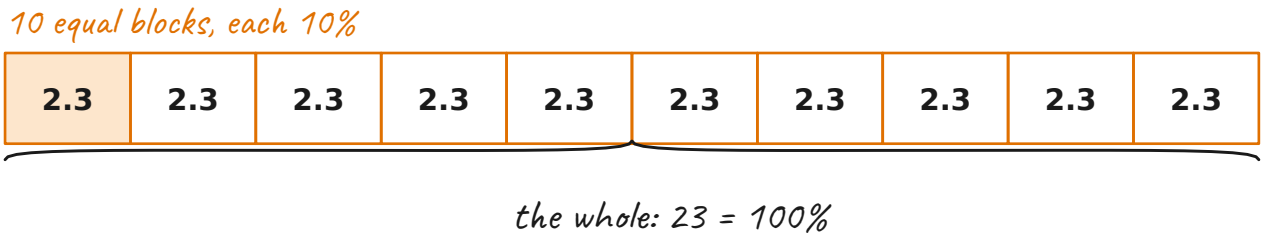
Finding 15% of 23: the 10% ladder

RULE: percentage of an amount

$x\%$ of $y = \frac{x}{100} \times y$ ("of" means $\times$).

Without a calculator: build it from 10% ($\div 10$), 5% (half of 10%) and 1% ($\div 100$).

You don't need a formula to begin with. **10%** is easy: it's a tenth, so divide by 10 (digits slide one place, Chapter 3). Build everything else from 10% and 1%:



$$10\% \text{ of } 23 = 2.3 \quad \leftarrow \text{divide by 10}$$

$$5\% \text{ of } 23 = 1.15 \quad \leftarrow \text{half of 10\%}$$

$$15\% \text{ of } 23 = 2.3 + 1.15 = 3.45 \quad \leftarrow \text{10\% + 5\%}$$

The one-line version from the lecture: "of" means multiply, and $x\%$ means $\frac{x}{100}$:

$$x\% \text{ of } y = \frac{x}{100} \times y$$

x is the percentage,
y is the whole amount

$$15\% \text{ of } 23 = 0.15 \times 23 = 3.45$$

And above 100%: 125% of 20 is $1.25 \times 20 = 25$. Bigger than 20, as it should be.

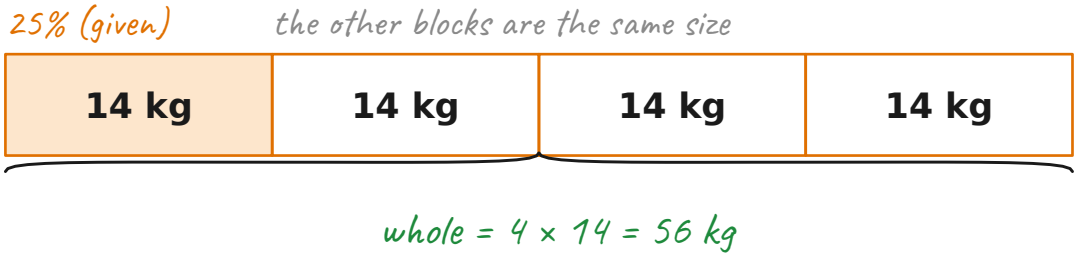
Working backwards: 25% of A is 14 kg

RULE: finding the whole from a percentage of it

If $p\%$ of the whole is N, then 1% is $N \div p$ and the whole (100%) is $\frac{N}{p} \times 100$.

Equivalently: whole $= N \div \frac{p}{100}$. E.g. 20% of a number is 9: $9 \div 0.2 = 45$.

You know a **part** and want the **whole**. Draw the whole as a bar. 25% is a quarter, so cut it into 4 blocks. One block is 14 kg:



The lecture's way (go through 1%, it always works, even for ugly percentages):

$$25\% \rightarrow 14 \quad \leftarrow \text{what we know}$$

$$1\% \rightarrow \frac{14}{25} \quad \leftarrow \div 25 \text{ on both sides}$$

$$100\% \rightarrow 100 \times \frac{14}{25} = 56 \quad \leftarrow \times 100 \text{ on both sides}$$

Common slip: 25% of A is 14, so A is 14 + 25% of 14? No! The 25% is of the **unknown whole**, not of 14. Always ask "percent of **what**?"

Going up and down: the multiplier

RULE: percentage increase and decrease (multipliers)

Increase by r%: new $=$ old $\times \left(1 + \frac{r}{100}\right)$. Up 12% means $\times 1.12$.

Decrease by r%: new $=$ old $\times \left(1 - \frac{r}{100}\right)$. Down 15% means $\times 0.85$.

Several changes in a row: multiply the multipliers. Up 10% then down 10% is $\times 1.1 \times 0.9 = \times 0.99$, a 1% fall overall (not 0%).

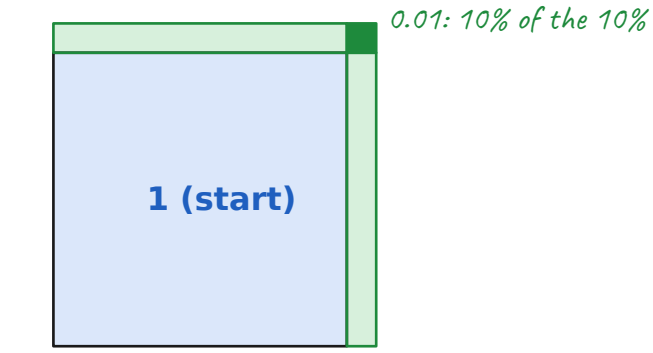
Increase 56.0 ml by 10%. The old amount is 100%. Adding 10% makes it **110%** of the old amount, so multiply by $\frac{110}{100} = 1.10$. That number is the **multiplier**.



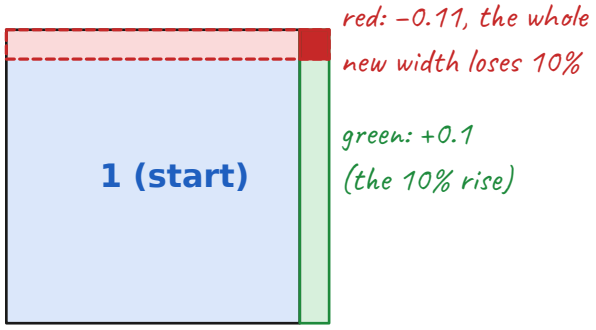
Up 10% then down 10% does NOT get you back to 56! The second 10% is 10% of a bigger number (61.6), so it takes away more (6.16) than was added (5.6).

(The lecture rounds 55.44 to 55.4.) Decreasing by 10% leaves **90%**, so the multiplier is 0.90, exactly as the rule box says: $1 - \frac{10}{100} = 0.9$.

Where did the missing 1% go? Draw it. Two multipliers in a row multiply, and multiplying is the area of a rectangle (Chapter 1). Let the starting amount be a 1 by 1 square:



up 10%, then up 10% again:
1 + 0.1 + 0.1 + 0.01 = 1.21



up 10%, then down 10%:
1 + 0.1 − 0.1 − 0.01 = 0.99

On the right, the 10% cut is taken off the **new, wider** amount, so it also slices off the little dark corner: 10% of the 10% that was added. That corner, 0.01, is the lost 1%. $56 \times 0.99 = 55.44$ ✓. On the left the same corner is a bonus: that's **interest on interest**, which Chapter 21 turns into compound growth.

A shop receipt: a £40 jacket in a 30% sale. Multiplier $\frac{70}{100} = 0.7$, so you pay $40 \times 0.7 =$ **£28**.

Percentage change: always compare to where you started

RULE: percentage change

Percentage change $= \dfrac{\text{new} - \text{original}}{\text{original}} \times 100.$

Divide by the **original** (the value before the change), never by the new value.

A positive answer is an increase; a negative answer is a decrease.

A price goes from £40 to £50. By what percentage did it rise? Take the **change** and compare it with the **original**:

$$\frac{50 - 40}{40} \times 100 = \frac{10}{40} \times 100 = 25\% \quad \leftarrow \text{a rise of 25\%}$$

$$\frac{40 - 50}{50} \times 100 = \frac{-10}{50} \times 100 = -20\% \quad \leftarrow \text{back down: only 20\% (and a minus sign means a fall)}$$

Same £10 gap, different percentages, because the "original" is different each time.

Reverse percentages: finding the original

A price **after** a percentage change is known, and you want the price **before** it.

PROCEDURE: reverse percentages

1. Write the multiplier for the change: $1 + \frac{r}{100}$ for an increase, $1 - \frac{r}{100}$ for a decrease.
2. new = original × multiplier, so **original = new ÷ multiplier**.
3. Check by applying the percentage change to your answer.

Example. After a 20% rise a jacket costs £60. What did it cost before?

$$\text{multiplier} = 1 + \frac{20}{100} = 1.2$$ ← a 20% increase

$$\text{original} = 60 \div 1.2 = £50$$ ← undo × 1.2 by ÷ 1.2

$$50 \times 1.2 = 60 \checkmark$$ ← check

Example. In a 15% sale a lamp costs £102. What was the full price?

$$102 \div 0.85 = £120$$ ← decrease of 15%: multiplier 0.85

$$120 \times 0.85 = 102 \checkmark$$ ← check

Example. A salary of £27 600 includes a 15% rise. What was it before?

$$27\,600 \div 1.15 = £24\,000$$

✗ Common mistakes

✗ $60 - 20\%$ of $60 = 48$ ✓ $60 \div 1.2 = 50$

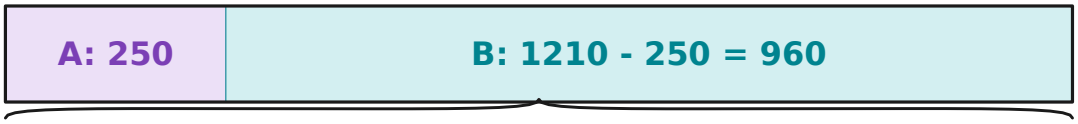
the 20% was of the ORIGINAL price, not of £60. Check: 48 + 20% of 48 = 57.60, not 60

✗ $102 + 15\%$ of $102 = 117.30$ ✓ $102 \div 0.85 = 120$

same mistake the other way: the 15% was taken off the unknown full price, not off £102

What percentage of the mixture is B?

250 mg of A is mixed with some B to make 1.21 g of mixture. What percentage is B? **First, same units**: 1.21 g is 1210 mg (1 g = 1000 mg).



whole mixture 1210 mg

$$\frac{960}{1210} \times 100 \approx 79.3\%$$

Part over whole, times 100 is the percentage every time.

Ratio: a recipe

RULE: what a ratio tells you

$a : b$ means: for every a parts of the first thing there are b parts of the second. **Order matters:** $1 : 4$ is not $4 : 1$.

There are $a + b$ parts altogether, so the first thing is $\frac{a}{a+b}$ of the total and the second is $\frac{b}{a+b}$. In squash $1 : 4$, squash is $\frac{1}{5}$ of the drink.

Simplify by dividing every part by the same number (ideally the HCF): $12 : 18 = 2 : 3$.

Same units first, then simplify: $50 \text{ cm} : 2 \text{ m} = 50 : 200 = 1 : 4$.

Squash bottles say **"1 part squash to 4 parts water"**. A "part" can be any size of cup: an egg-cup or a bucket. The ratio only tells you the **shape of the recipe**, not the amount. We write it **1 : 4**, read "one **to** four".



egg-cups: 1 + 4



mugs: 1 + 4



jugs: 1 + 4

Three very different amounts, all the same taste: every one is 1 : 4.

Double the parts, same ratio: 2 : 8 tastes identical to 1 : 4. So ratios **simplify** exactly like fractions (Chapter 2): divide every number by the same thing.

$$2 : 8 \;=\; 1 : 4 \qquad 260 : 100 : 20 \;=\; 13 : 5 : 1$$

÷ 2 ÷ 20 (the HCF): a ratio can have 3 parts!

Order matters: squash : water 1 : 4 is not water : squash 1 : 4. That second one would be very strong squash!

Mixed units. Simplify $1.2\ \text{kg} : 300\ \text{g}$.

$$1.2\ \text{kg} = 1200\ \text{g}$$ ← 1. same units (grams)

$$1200 : 300 = 4 : 1$$ ← 2. ÷ 300 (the HCF)

Reading the lecture's notation: a : b :: 1 : 72

A model aeroplane: 1 cm on the model stands for 72 cm on the real plane. The scale is **1 : 72**. If the model is a cm long and the real plane is b cm long:

$$a : b \;::\; 1 : 72$$

say: "a is to b as 1 is to 72"
the :: means "is in the same ratio as"

Every ratio statement can be rewritten as a fraction statement. These all say the same thing (the lecture lists them):

$$\frac{a}{b} = \frac{1}{72} \qquad b = 72a \qquad a = \frac{b}{72}$$

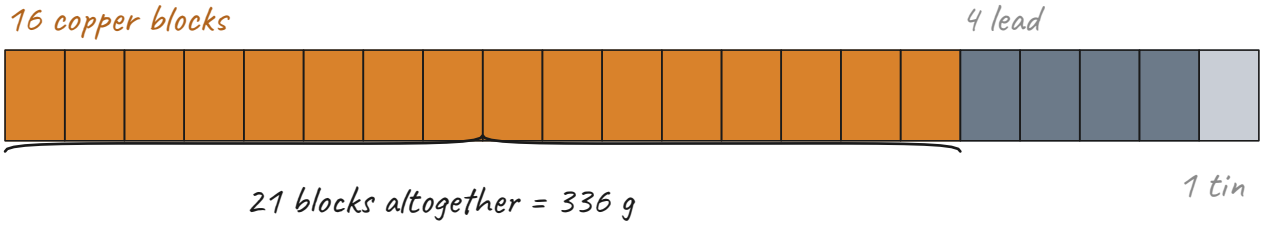
Say it in words: the real thing is 72 times longer than the model; the model is one 72nd of the real thing. A 30 cm model means a $30 \times 72 = 2160$ cm (21.6 m) plane.

Sharing in a ratio: count the parts

PROCEDURE: sharing an amount in a ratio

1. **Add the parts** to get the total number of parts: for $a : b : c$ that is $a + b + c$.
2. **Divide** the total amount by the number of parts: this is the size of **one part**.
3. **Multiply** one part by each number in the ratio.
4. Check that the shares add back up to the total.

Lecture 3, Example 3. 336 g of alloy is copper, lead and tin in the ratio 16 : 4 : 1. How much of each? The lecture's advice: **count the parts**.



$$16 + 4 + 1 = 21 \text{ parts}$$ ← 1. count the parts

$$336 \div 21 = 16 \text{ g per part}$$ ← 2. how heavy is one part?

$$16 \times 16 = 256, \quad 4 \times 16 = 64, \quad 1 \times 16 = 16$$

3. each metal

$$256 + 64 + 16 = 336 \checkmark$$ ← 4. check they add back up

Then 4 g copper, 36 g lead and 4 g tin are added: 260 g, 100 g and 20 g (380 g in all). The new ratio is 260 : 100 : 20, which simplifies (÷ 20) to **13 : 5 : 1**.

PROCEDURE: when you are given one share or the difference, not the total

1. Whatever you are told, first turn it into **"so many parts = this amount"**.
2. Given one share: that share is its own number of parts. Given a difference: subtract the parts.
3. Divide to find one part, then multiply up to whatever is asked for.

Example (one share given). Money is shared A : B = 3 : 5. B gets £40. Find A's share and the total.

$$5 \text{ parts} = £40 \;\Rightarrow\; 1 \text{ part} = £8$$ ← *B is 5 parts*

$$A = 3 \times 8 = £24, \quad \text{total} = 8 \times 8 = £64$$

3 + 5 = 8 parts in all

Example (difference given). A : B = 5 : 3, and A gets £20 more than B. How much does each get?

$$5 - 3 = 2 \text{ parts} = £20 \;\Rightarrow\; 1 \text{ part} = £10$$

the difference is 2 parts

$$A = 5 \times 10 = £50, \quad B = 3 \times 10 = £30$$

check: 50 − 30 = 20

Ratios as percentages (and back)

RULE: converting between ratios and percentages

Ratio to percentage: each part's percentage $= \dfrac{\text{its number}}{\text{sum of all the numbers}} \times 100.$

Percentages to ratio: write the percentages as a ratio and simplify: $30\% : 70\% = 30 : 70 = 3 : 7.$

The percentages of all the parts add up to 100% (check, allowing for rounding).

Lecture 3. The alloy is copper : lead : tin $= 16 : 4 : 1$. What percentage of it is each metal?

$$16 + 4 + 1 = 21 \text{ parts}$$ ← *the whole is 21 parts*

$$\text{copper} = \frac{16}{21} \times 100 \approx 76.2\%$$

$$\text{lead} = \frac{4}{21} \times 100 \approx 19.0\%$$

$$\text{tin} = \frac{1}{21} \times 100 \approx 4.8\%$$ ← *76.2 + 19.0 + 4.8 = 100 ✓*

Back again. A solution is 35% acid and the rest water. Acid : water $= 35 : 65 = 7 : 13$ (÷ 5).

The "one of them" table (unitary method)

Lecture 3, Example 2. 2.504 g of copper oxide contains 2 g of copper. How much copper is in 0.939 g of copper oxide? Make a two-column table and always pass through **one**:

copper oxide	copper
2.504 g	2 g
1 g	$\dfrac{2}{2.504}$ g
0.939 g	$0.939 \times \dfrac{2}{2.504} = 0.75$ g

÷ 2.504 on both sides

× 0.939 on both sides

Whatever you do to one column, do to the other. That keeps the ratio the same, which is what **proportion** means: the two amounts grow and shrink together.

Example 1 with the same table: speeds U and V are in the ratio 8 : 3, and U = 5 m/s. Find V.

$$8 \text{ parts} \to 5$$ ← U is 8 parts

$$1 \text{ part} \to \frac{5}{8}$$ ← ÷ 8

$$3 \text{ parts} \to 3 \times \frac{5}{8} = \frac{15}{8} = 1.875$$ ← × 3: V = 1.875 m/s

The slide does the same examples by **cross-multiplying**; that rule is stated in the next section.

Proportions: cross-multiplying, and inverse proportion

RULE: cross-multiplication

$a : b = c : d$ (also written $a : b :: c : d$) means $\dfrac{a}{b} = \dfrac{c}{d}$.

If $\dfrac{a}{b} = \dfrac{c}{d}$ **then** $a \times d = b \times c$: multiply each top by the **other** bottom.

Why: multiply both sides by b and by d; the bottoms cancel. (Chapter 15 uses this for equations.)

Then divide to get the unknown on its own.

Lecture 3, Example 1. $U : V = 8 : 3$ and $U = 5$ m/s. Find V.

$$\frac{5}{V} = \frac{8}{3}$$ ← U/V = 8/3 with U = 5

$$5 \times 3 = 8 \times V \;\Rightarrow\; 15 = 8V$$ ← cross-multiply

$$V = \frac{15}{8} = 1.875 \text{ m/s}$$ ← ÷ 8

Lecture 3, Example 2a. 2.504 g of copper oxide contains 2 g of copper. How much copper is in 0.939 g? Call it x g.

$$\frac{2}{x} = \frac{2.504}{0.939}$$ ← copper 2 : x matches oxide 2.504 : 0.939

$$2 \times 0.939 = 2.504x \;\Rightarrow\; x = \frac{1.878}{2.504} = 0.75 \text{ g}$$

cross-multiply, ÷ 2.504

Missing term in a proportion. $3 : 8 = x : 20$

$$\frac{3}{8} = \frac{x}{20} \Rightarrow 8x = 60 \Rightarrow x = 7.5$$ ← 3 × 20 = 8 × x

RULE: direct and inverse proportion

Direct proportion: both amounts go up together; double one and the other doubles. The **ratio** $\frac{y}{x}$ stays the same (the unitary method and cross-multiplying work).

Inverse proportion: one goes up as the other goes down; double one and the other **halves**. The **product** $x \times y$ stays the same.

Typical inverse case: more workers, fewer days; faster speed, less time.

Example (inverse). 4 people paint a fence in 6 hours. How long would 3 people take (same speed each)?

$$4 \times 6 = 24 \text{ person-hours}$$ ← *the product stays fixed*

$$24 \div 3 = 8 \text{ hours}$$ ← *fewer people, more time*

Not $6 \times \frac{3}{4} = 4.5$ hours: that would be treating it as direct proportion. Ask "does more of one mean more or less of the other?" first.

Ugly ratios: the lecture's tidy-up trick

Measured ratios are rarely neat, e.g. 262 : 96.7 : 18.1. To make one easy to read:

$$262 : 96.7 : 18.1$$ ← *the raw ratio*

$$2620 : 967 : 181$$ ← *1. × 10 to remove the decimals*

$$14.48 : 5.34 : 1$$ ← *2. ÷ the smallest (181): now it's "per 1 of the last"*

$$1448 : 534 : 100$$ ← *3. × 100 (optional, whole numbers)*

Step 2 is the useful one: "14.48 of the first for every 1 of the last".

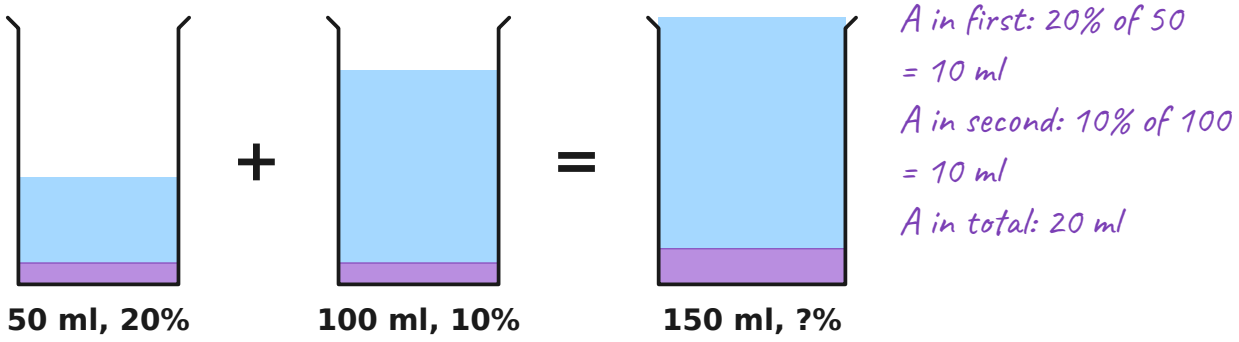
Mixing solutions: beakers

A "20% solution of A" means 20% of the liquid is chemical A and the rest is water. This percentage is called the **concentration** or **strength**.

PROCEDURE: mixing solutions

1. Amount of A in a solution = concentration × volume (e.g. 20% of 50 ml = 10 ml).
2. Add the amounts of A, and add the volumes.
3. New concentration $= \dfrac{\text{total A}}{\text{total volume}} \times 100$. **Never average the percentages.**

Lecture 3, Mixing 1: 50 ml of 20% solution is added to 100 ml of 10% solution. How strong is the result? **Track the A, not the percentages**:



A in first: 20% of 50 = 10 ml
A in second: 10% of 100 = 10 ml
A in total: 20 ml

purple = chemical A, blue = water. (Real chemistry check from the slides: 100 ml alcohol + 100 ml water gives only about 190 ml!)

$$\frac{20}{150} \times 100 = 13.\dot{3}\% \quad \longleftarrow \text{A over whole, × 100}$$

The tempting wrong answer is to average: (20 + 10) ÷ 2 = 15%. That ignores that there is twice as much of the weaker one, so the answer is pulled towards 10%.

Find the unknown

RULE: which rule do I need?

Value **after** a % change known, original wanted: **divide** by the multiplier.

One share (or the difference between shares) known, total wanted: turn it into "so many parts $=$ amount", find one part, multiply up.

Three of the four numbers in $a : b = c : d$ known: cross-multiply, then divide.

Part and percentage known, whole wanted: whole $=$ part $\div \frac{p}{100}$.

Example. A car loses 30% of its value and is now worth £8 400. What was it worth?

$$8400 \div 0.7 = £12\,000$$ ← decrease 30%: multiplier 0.7

Example. Sand and cement are mixed 5 : 2. 14 kg of cement is used. How much mortar is made?

$$2 \text{ parts} = 14 \text{ kg} \Rightarrow 1 \text{ part} = 7 \text{ kg}$$ ← cement is 2 parts

$$7 \times 7 = 49 \text{ kg}$$ ← 5 + 2 = 7 parts in total

Example. $\frac{x}{12} = \frac{5}{8}$

$$8x = 60 \Rightarrow x = 7.5$$ ← cross-multiply: x × 8 = 12 × 5

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Per cent	$x\% = \dfrac{x}{100}$ % → decimal: ÷ 100; decimal or fraction → %: × 100
Percentage of	$x\%$ of $y = \dfrac{x}{100} \times y$
Whole from a part	$\text{whole} = \text{part} \div \dfrac{p}{100}$
Increase / decrease by r%	$\text{new} = \text{old} \times \left(1 \pm \dfrac{r}{100}\right)$ + for an increase, − for a decrease; repeated changes: multiply the multipliers
Percentage change	$\dfrac{\text{new} - \text{old}}{\text{old}} \times 100$ divide by the original (old) value
Reverse percentage	$\text{original} = \dfrac{\text{new}}{1 \pm \frac{r}{100}}$ divide by the multiplier; **not** new − r% of new
Ratio as fractions	$a : b \Rightarrow \dfrac{a}{a+b}$ and $\dfrac{b}{a+b}$ of the total
Simplify a ratio	same units first, then divide every part by the HCF
Share in a ratio	add the parts, divide the total, multiply each part
One share or difference	"so many parts = amount", then one part, then multiply
Ratio to percentage	$\dfrac{\text{part}}{\text{sum of parts}} \times 100$
Cross-multiplication	$\dfrac{a}{b} = \dfrac{c}{d} \Rightarrow ad = bc$ direct proportion: the ratio stays the same
Inverse proportion	$x \times y = \text{constant}$
Mixing	$\dfrac{\text{total A}}{\text{total volume}} \times 100$ never average the percentages

Recipe cards: your turn

Answers under each group, small and grey. Draw a bar or a table for every one!

Harder mixtures ("how much 15% solution must be added to...?" and the two-solution "compounds" problems) need an equation with a letter in it. They're solved step by step in Chapter 15.

Percentages

1. Write 35% as a decimal and a fraction
2. Write 0.07 as a percentage
3. Find 20% of 65
4. Find 17.5% of 80
5. 30% of a number is 12. Find the number

1. 0.35 = 7/20 2. 7% 3. 13 4. 14 5. 40

Up, down and change

6. Increase 250 by 12%
7. Decrease 80 by 15%
8. A price falls from £60 to £45. Percentage change?
9. What single multiplier is +20% then -20%?

6. 280 7. 68 8. -25% 9. 1.2 × 0.8 = 0.96 (a 4% fall)

Ratio

10. Simplify 45 : 60
11. Share £84 in the ratio 3 : 4
12. Squash 2 : 7. 450 ml of drink: how much squash?
13. Map scale 1 : 25,000. 4 cm on the map is how many km?

10. 3 : 4 11. £36 and £48 12. 100 ml 13. 1 km

Mixtures

14. 200 ml of 30% solution + 300 ml of 5% solution. Strength?

14. (60 + 15)/500 = 15%

Ratio rules

15. Simplify 50 cm : 2 m
16. Write the ratio 9 : 3 : 3 as percentages
17. A solution is 40% acid, the rest water. Write acid : water
18. 6 workers take 10 days. How long would 4 workers take?
19. A : B = 7 : 2 and A gets £35 more than B. Find each share

15. 1 : 4 (50 : 200) 16. 60%, 20%, 20% (9/15, 3/15, 3/15) 17. 2 : 3 (40 : 60) 18. 15 days (inverse: 6 × 10 = 60, ÷ 4) 19. A £49, B £14 (5 parts = £35, 1 part = £7)

Find the unknown

20. After a 10% rise a price is £55. What was it before?
21. After a 25% discount a coat costs £45. What was the full price?
22. Boys : girls = 4 : 5 and there are 20 girls. How many pupils altogether?
23. x : 15 = 4 : 6. Find x
24. 2.5 / x = 5 / 8. Find x
25. 12% of a number is 18. Find the number

20. £50 (55 ÷ 1.1) 21. £60 (45 ÷ 0.75) 22. 36 (5 parts = 20, 1 part = 4, 9 parts) 23. 10 (6x = 15 × 4 = 60) 24. 4 (5x = 2.5 × 8 = 20) 25. 150 (18 ÷ 0.12)